COHOMOLOGY OF A COLIMIT OF SHEAVES OVER A COLIMIT OF SITES AS THE COLIMIT OF THE COHOMOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all degrees; the reduction is modelled on the companion result of the same chapter and the Cech computation is carried out here.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The inverse system of sites and the sheaves attached to it. Same chapter, the statement of lemma-colim-sites-injective delivered verbatim as a context span in this card (source lines 2854-2875), fixing the cofinal index category, the sites, the sheaves and their compatibility maps. Supplies every object appearing in the counted statement and the embedding used in the proof.
- Cech cohomology vanishes for the members of the system and filtered colimits of acyclic complexes are acyclic. Same chapter, the vanishing of Cech cohomology of injective sheaves, the criterion computing cohomology by Cech cohomology over a cofinal collection of coverings, and the exactness of filtered colimits. Supplies the three ingredients of the Cech computation.

Source reference. Original reference 0A34: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2858–2858. Navigation pointer invoked at proof line 2858; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09YL: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2860–2860. Navigation pointer invoked at proof

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line 2860; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A35: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2861–2861. Navigation pointer invoked at proof line 2861; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09YN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2917–2917. Navigation pointer invoked at proof line 2917; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0739: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2518–2518. Navigation pointer invoked at proof line 2923; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1565–1565. Navigation pointer invoked at proof line 2932; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1334–1334. Navigation pointer invoked at proof line 2942; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00DB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2943–2943. Navigation pointer invoked at proof line 2943; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered index category and let $(\mathcal{C}_i, f_a)$ be an inverse system of sites over $\mathcal{I}$ as in Sites, Situation 0A34. Set $\mathcal{C} = \operatorname{colim} \mathcal{C}_i$ as in Sites, Lemmas 09YL and 0A35. Moreover, assume given*

(1) *an abelian sheaf $\mathcal{F}_i$ on $\mathcal{C}_i$ for all $i \in \mathrm{Ob}(\mathcal{I})$,*
(2) *for $a : j \to i$ a map $\varphi_a : f_a^{-1}\mathcal{F}_i \to \mathcal{F}_j$ of abelian sheaves on $\mathcal{C}_j$*

such that $\varphi_c = \varphi_b \circ f_b^{-1}\varphi_a$ whenever $c = a \circ b$. Then there exists a map of systems $(\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ such that $\mathcal{F}_i \to \mathcal{G}_i$ is injective and $\mathcal{G}_i$ is an injective abelian sheaf.

Lemma 0.2. *In the situation of Lemma 0.1 set $\mathcal{F} = \operatorname{colim} f_i^{-1}\mathcal{F}_i$. Let $i \in \mathrm{Ob}(\mathcal{I})$, $X_i \in Ob(\mathcal{C}_i)$. Then*

$$\operatorname{colim}_{a:j\to i} H^p(u_a(X_i), \mathcal{F}_j) = H^p(u_i(X_i), \mathcal{F})$$

for all $p \geq 0$.

Proof. The case $p = 0$ is Sites, Lemma 09YN.

Choose $(\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ as in Lemma 0.1. Arguing exactly as in the proof of Lemma 0739 we see that it suffices to prove that $H^p(X, \operatorname{colim} f_i^{-1}\mathcal{G}_i) = 0$ for $p > 0$.

Set $\mathcal{G} = \operatorname{colim} f_i^{-1}\mathcal{G}_i$. To show vanishing of cohomology of $\mathcal{G}$ on every object of $\mathcal{C}$ we show that the Čech cohomology of $\mathcal{G}$ for any covering $\mathcal{U}$ of $\mathcal{C}$ is zero (Lemma 03F9). The covering $\mathcal{U}$ comes from a covering $\mathcal{U}_i$ of $\mathcal{C}_i$ for some i. We have

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{G}) = \operatorname{colim}_{a:j\to i} \check{\mathcal{C}}^\bullet(u_a(\mathcal{U}_i), \mathcal{G}_j)$$

by the case $p = 0$. The right hand side is acyclic in positive degrees as a filtered colimit of acyclic complexes by Lemma 03AW. See Algebra, Lemma 00DB. □